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1 import numpy as np
2 from scipy.optimize import nnls
3
4 species = [
5     'IB',          # 0
6     'MAC',         # 1
7     'CO',          # 2
8     'MAA',         # 3
9     'Acetic acid', # 4
10    'Acetone',      # 5
11    'Acetaldehyde', # 6
12    'Acrolein',     # 7
13    'Formaldehyde', # 8
14    'Acrylic acid', # 9
15    'O2',           #10
16    'N2',           #11
17    'CO2',          #12
18    'H2O',          #13
19    'Formic acid'   #14
20 ]
21
22 F_in = np.array([
23     41.75,  # IB
24     0.00,   # MAC
25     0.00,   # CO
26     0.00,   # MAA
27     0.00,   # Acetic acid
28     0.00,   # Acetone
29     0.00,   # Acetaldehyde
30     0.00,   # Acrolein
31     0.00,   # Formaldehyde
32     0.00,   # Acrylic acid
33     83.50,  # O2
34     296.05, # N2
35     0.00,   # CO2
36     62.63,  # H2O
37     0.00    # Formic acid
38 ])
39
40 F_out = np.array([
41     0.17,  # IB
42     34.47, # MAC
43     5.12,  # CO
44     1.45,  # MAA
45     1.45,  # Acetic acid
46     0.83,  # Acetone
47     1.45,  # Acetaldehyde
48     0.28,  # Acrolein
49     0.21,  # Formaldehyde
50     0.21,  # Acrylic acid
51     26.35, # O2
52     296.05, # N2
53     7.56,  # CO2
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53         113.37, # CO2
54         113.37, # H2O
55         0.00     # Formic acid
56 ])
57
58 DeltaF = F_out - F_in
59 nu = np.zeros((14, 15))
60 # r1: IB + O2 → MAC + H2O
61 nu[0,0] = -1      # IB
62 nu[0,10] = -1     # O2
63 nu[0,1] = +1     # MAC
64 nu[0,13] = +1    # H2O
65
66 # r2: MAC + 0.5 O2 → MAA
67 nu[1,1] = -1     # MAC
68 nu[1,10] = -0.5  # O2
69 nu[1,3] = +1     # MAA
70
71 # r3: MAC + 4O2 → Acetic acid + CO2 + 2H2O
72 nu[2,1] = -1
73 nu[2,10] = -4
74 nu[2,4] = +1     # Acetic acid
75 nu[2,12] = +1    # CO2
76 nu[2,13] = +2    # H2O
77
78 # r4: MAC + 3O2 → Acrylic acid + H2O
79 nu[3,1] = -1
80 nu[3,10] = -3
81 nu[3,9] = +1     # Acrylic acid
82 nu[3,13] = +1
83
84 # r5: MAC + 2.5O2 → Acrolein + CO2 + H2O
85 nu[4,1] = -1
86 nu[4,10] = -2.5
87 nu[4,7] = +1     # Acrolein
88 nu[4,12] = +1
89 nu[4,13] = +1
90
91 # r6: MAC + 3O2 → Acetaldehyde + 2CO + H2O
92 nu[5,1] = -1
93 nu[5,10] = -3
94 nu[5,6] = +1     # Acetaldehyde
95 nu[5,2] = +2     # CO
96 nu[5,13] = +1
97
98 # r7: IB + 6O2 → 4CO2 + 4H2O
99 nu[6,0] = -1
100 nu[6,10] = -6
101 nu[6,12] = +4
102 nu[6,13] = +4
103
104 # r8: IB + 4O2 → 4CO + 4H2O
105 nu[7,0] = -1
106 nu[7,10] = -4
107 nu[7,2] = +4

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108 nu[7,13] = +4
109
110 # r9: MAC + 1.5 O2 → Acrolein + CO2 + H2O
111 nu[8,1] = -1
112 nu[8,10] = -1.5
113 nu[8,7] = +1
114 nu[8,12] = +1
115 nu[8,13] = +1
116
117 # r10: MAC + 5O2 → 4CO2 + 3H2O
118 nu[9,1] = -1
119 nu[9,10] = -5
120 nu[9,12] = +4
121 nu[9,13] = +3
122
123 # r11: MAC + 3O2 → 4CO + 3H2O
124 nu[10,1] = -1
125 nu[10,10] = -3
126 nu[10,2] = +4
127 nu[10,13] = +3
128
129 # r12: IB + O2 → Acetone + Formaldehyde
130 nu[11,0] = -1
131 nu[11,10] = -1
132 nu[11,5] = +1      # Acetone
133 nu[11,8] = +1      # Formaldehyde
134
135 # r13: Formaldehyde + 0.5O2 → CO
136 nu[12,8] = -1
137 nu[12,10] = -0.5
138 nu[12,2] = +1
139
140 # r14: Formaldehyde + O2 → CO2 + H2O
141 nu[13,8] = -1
142 nu[13,10] = -1
143 nu[13,12] = +1
144 nu[13,13] = +1
145
146 # NNLS solution
147 xi, residuals = nnls(nu.T, DeltaF)
148
149 print("Extents of reaction (kmol/h):")
150 for i, ext in enumerate(xi, start=1):
151     print(f"r{i:2d} = {ext:10.4f}")
152
153 DeltaF_pred = nu.T @ xi
154 print("\nChange in molar flow (kmol/h):")
155 print(f"{'Species':15} {'Actual':>10} {'Predicted':>12} {'Diffe")
156 for name, act, pred in zip(species, DeltaF, DeltaF_pred):
157     print(f"{name:15} {act:10.3f} {pred:12.3f} {act-pred:12.3f}")
158
159 print(f"\nSum of squared residuals: {residuals:.3f}")

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Extents of reaction (kmol/h):

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r 1 = 38.0162
r 2 = 1.4648
r 3 = 0.7133
r 4 = 0.0000
r 5 = 0.0000
r 6 = 0.6670
r 7 = 1.3210
r 8 = 0.9343
r 9 = 0.4387
r10 = 0.0000
r11 = 0.0000
r12 = 0.8936
r13 = 0.0000
r14 = 0.8275

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Change in molar flow (kmol/h):

Species	Actual	Predicted	Difference
IB	-41.580	-41.165	-0.415
MAC	34.470	34.732	-0.262
CO	5.120	5.071	0.049
MAA	1.450	1.465	-0.015
Acetic acid	1.450	0.713	0.737
Acetone	0.830	0.894	-0.064
Acetaldehyde	1.450	0.667	0.783
Acrolein	0.280	0.439	-0.159
Formaldehyde	0.210	0.066	0.144
Acrylic acid	0.210	0.000	0.210
O2	-57.150	-57.645	0.495
N2	0.000	0.000	0.000
CO2	7.560	7.264	0.296
H2O	50.740	50.397	0.343
Formic acid	0.000	0.000	0.000

Sum of squared residuals: 1.394

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1 import numpy as np
2 from scipy.optimize import nnls
3
4 species = [
5     'IB',          # 0
6     'MAC',         # 1
7     'MAA',         # 2
8     'CO',          # 3
9     'CO2',         # 4
10    'Acetic acid',  # 5
11    'Acetone',      # 6
12    'Acetaldehyde', # 7
13    'Acrolein',     # 8
14    'Formaldehyde', # 9
15    'Acrylic acid', # 10
16    'O2',           # 11
17    'N2',           # 12
18    'H2O',          # 13
19    'Formic acid'   # 14
20 ]
21
22

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22 F_in = np.array([
23     0.17,      # IB
24     34.47,     # MAC
25     1.45,      # MAA
26     5.12,      # CO
27     7.56,      # CO2
28     1.45,      # Acetic acid
29     0.83,      # Acetone
30     1.45,      # Acetaldehyde
31     0.28,      # Acrolein
32     0.21,      # Formaldehyde
33     0.21,      # Acrylic acid
34     176.35,    # O2
35     827.86,    # N2
36     113.37,    # H2O
37     0.00       # Formic acid
38 ])
39
40 F_out = np.array([
41     0.00,      # IB
42     7.58,      # MAC
43     24.70,     # MAA
44     10.83,     # CO
45     36.00,     # CO2
46     3.36,      # Acetic acid
47     1.30,      # Acetone
48     2.27,      # Acetaldehyde
49     0.43,      # Acrolein
50     0.32,      # Formaldehyde
51     0.32,      # Acrylic acid
52     70.41,     # O2
53     827.86,    # N2
54     205.64,    # H2O
55     0.00       # Formic acid
56 ])
57
58 DeltaF = F_out - F_in
59 nu = np.zeros((14, 15))
60
61 # r1: IB + O2 -> MAC + H2O
62 nu[0,0] = -1
63 nu[0,11] = -1
64 nu[0,1] = +1
65 nu[0,13] = +1
66
67 # r2: IB + 1.5 O2 -> MAA + H2O
68 nu[1,0] = -1
69 nu[1,11] = -1.5
70 nu[1,2] = +1
71 nu[1,13] = +1
72
73 # r3: IB + 5O2 -> Acetic acid + CO2 + 3H2O
74 nu[2,0] = -1
75 nu[2,11] = -5
76 nu[2,5] = +1

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76 nu[2,5] = +1
77 nu[2,4] = +1
78 nu[2,13] = +3
79
80 # r4: IB + 2.5O2 -> Acrolein + CO2 + 2H2O
81 nu[3,0] = -1
82 nu[3,11] = -2.5
83 nu[3,8] = +1
84 nu[3,4] = +1
85 nu[3,13] = +2
86
87 # r5: IB + 3O2 -> Acetaldehyde + 2CO + 2H2O
88 nu[4,0] = -1
89 nu[4,11] = -3
90 nu[4,7] = +1
91 nu[4,3] = +2
92 nu[4,13] = +2
93
94 # r6: IB + O2 -> Acetone + Formaldehyde
95 nu[5,0] = -1
96 nu[5,11] = -1
97 nu[5,6] = +1
98 nu[5,9] = +1
99
100 # r7: IB + 3O2 -> Acrylic acid + 2H2O
101 nu[6,0] = -1
102 nu[6,11] = -3
103 nu[6,10] = +1
104 nu[6,13] = +2
105
106 # r8: MAC + 0.5 O2 -> MAA
107 nu[7,1] = -1
108 nu[7,11] = -0.5
109 nu[7,2] = +1
110
111 # r9: MAC + 5O2 -> 4CO2 + 3H2O
112 nu[8,1] = -1
113 nu[8,11] = -5
114 nu[8,4] = +4
115 nu[8,13] = +3
116
117 # r10: MAC + 4O2 -> 4CO + 4H2O
118 nu[9,1] = -1
119 nu[9,11] = -4
120 nu[9,3] = +4
121 nu[9,13] = +4
122
123 # r11: MAC + 3.5O2 -> Acetic acid + CO2 + 2H2O
124 nu[10,1] = -1
125 nu[10,11] = -3.5
126 nu[10,5] = +1
127 nu[10,4] = +1
128 nu[10,13] = +2
129
130 # r12: MAC + 3O2 -> Acrolein + CO2 + H2O
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130 # r12: MAC + 2O2 -> Acrolein + CO2 + H2O
131 nu[11,1] = -1
132 nu[11,11] = -2
133 nu[11,8] = +1
134 nu[11,4] = +1
135 nu[11,13] = +1
136
137 # r13: MAC + 2.5O2 -> Acetaldehyde + 2CO + H2O
138 nu[12,1] = -1
139 nu[12,11] = -2.5
140 nu[12,7] = +1
141 nu[12,3] = +2
142 nu[12,13] = +1
143
144 # r14: MAC + 1.5O2 -> Acetone + Formaldehyde + H2O
145 nu[13,1] = -1
146 nu[13,11] = -1.5
147 nu[13,6] = +1
148 nu[13,9] = +1
149 nu[13,13] = +1
150
151 xi, residuals = nnls(nu.T, DeltaF)
152
153 print("Extents of reaction (kmol/h):")
154 reactions = [
155     "IB + O2 -> MAC + H2O",
156     "IB + 1.5 O2 -> MAA + H2O",
157     "IB + 5O2 -> Acetic acid + CO2 + 3H2O",
158     "IB + 2.5O2 -> Acrolein + CO2 + 2H2O",
159     "IB + 3O2 -> Acetaldehyde + 2CO + 2H2O",
160     "IB + O2 -> Acetone + Formaldehyde",
161     "IB + 3O2 -> Acrylic acid + 2H2O",
162     "MAC + 0.5 O2 -> MAA",
163     "MAC + 5O2 -> 4CO2 + 3H2O",
164     "MAC + 4O2 -> 4CO + 4H2O",
165     "MAC + 3.5O2 -> Acetic acid + CO2 + 2H2O",
166     "MAC + 2O2 -> Acrolein + CO2 + H2O",
167     "MAC + 2.5O2 -> Acetaldehyde + 2CO + H2O",
168     "MAC + 1.5O2 -> Acetone + Formaldehyde + H2O"
169 ]
170
171 for i, (ext, rxn) in enumerate(zip(xi, reactions), start=1):
172     print(f"r{i:2d} = {ext:10.4f} # {rxn}")
173
174 DeltaF_pred = nu.T @ xi
175 print("\nChange in molar flow (kmol/h):")
176 print(f"{'Species':15} {'Actual':>10} {'Predicted':>12} {'Diff'
177 for name, act, pred in zip(species, DeltaF, DeltaF_pred):
178     print(f"{name:15} {act:10.3f} {pred:12.3f} {act-pred:12.3f}
179
180 print(f"\nSum of squared residuals: {residuals:.3f}")

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Extents of reaction (kmol/h):
r 1 =      0.0000 # IB + O2 -> MAC + H2O
r 2 =     10.1891 # IB + 1.5 O2 -> MAA + H2O

```